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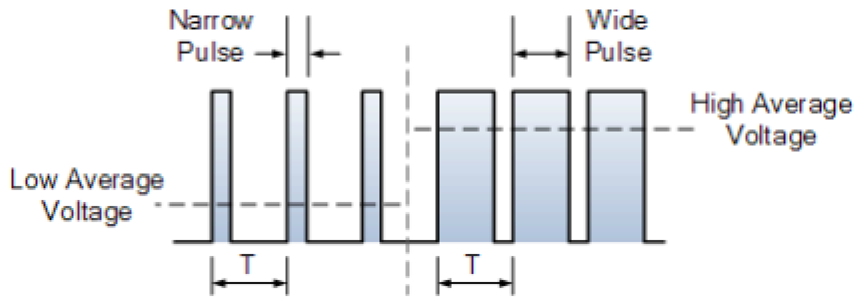
LAB 11

Pulse width and Motors Lab

Introduction

In previous labs, we learned that the speed of a motor can be controlled with a few techniques (that tend to be in-efficient). For example, technically you can control the speed of the motor by putting a large variable resistor infront of it (rheostate) in series to control the voltage, but that generates a lot of heat and wastes power. We also learned from previous labs that when we increase the voltage that increases the speed, and likewise decreasing the voltage also decreases the speed. The most efficient way to control the speed of a motor us to use PWM- pulse width modification. Pulse width modification controls the voltage by controlling the average pulse width. It works by driving the motor with on/ off pulses and varying duty circle (the fraction of time that the output voltage is on compared to when it is off) while maintaining a consistent frequency. The power applied to the motor can be controlled by varying the width of these applied pulses and thereby varying the average DC voltage applied to the motors terminals. By changing the time of these pulses the speed of the motor is controlled- the longer the pulse is on, the faster the motor will rotate, and the shorter the pulse is on the slower the motor will rotate. The wider the pulse width, the more average voltage applied to the motor terminals, the stronger the magnetic flux inside the armature windings is and the faster the motor will rotate.

In this lab, we explore a hybrid approach that combines traditional PWM with Pulse Position Modulation (PPM). This technique allows for more precise control by incorporating not only the width but also the timing (or position) of the pulses. Such hybrid modulation is particularly useful in applications requiring synchronized signals or in systems where multiple signals are transmitted over a single communication line.



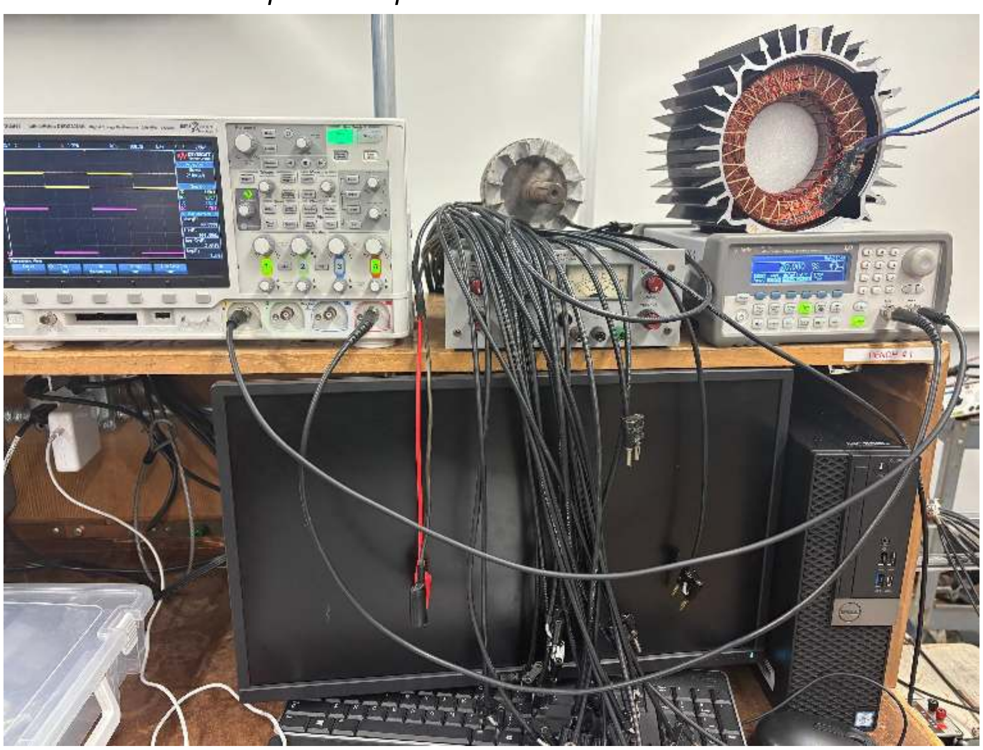
IMG 1. How PWM impacts the voltage

OBJECTIVES

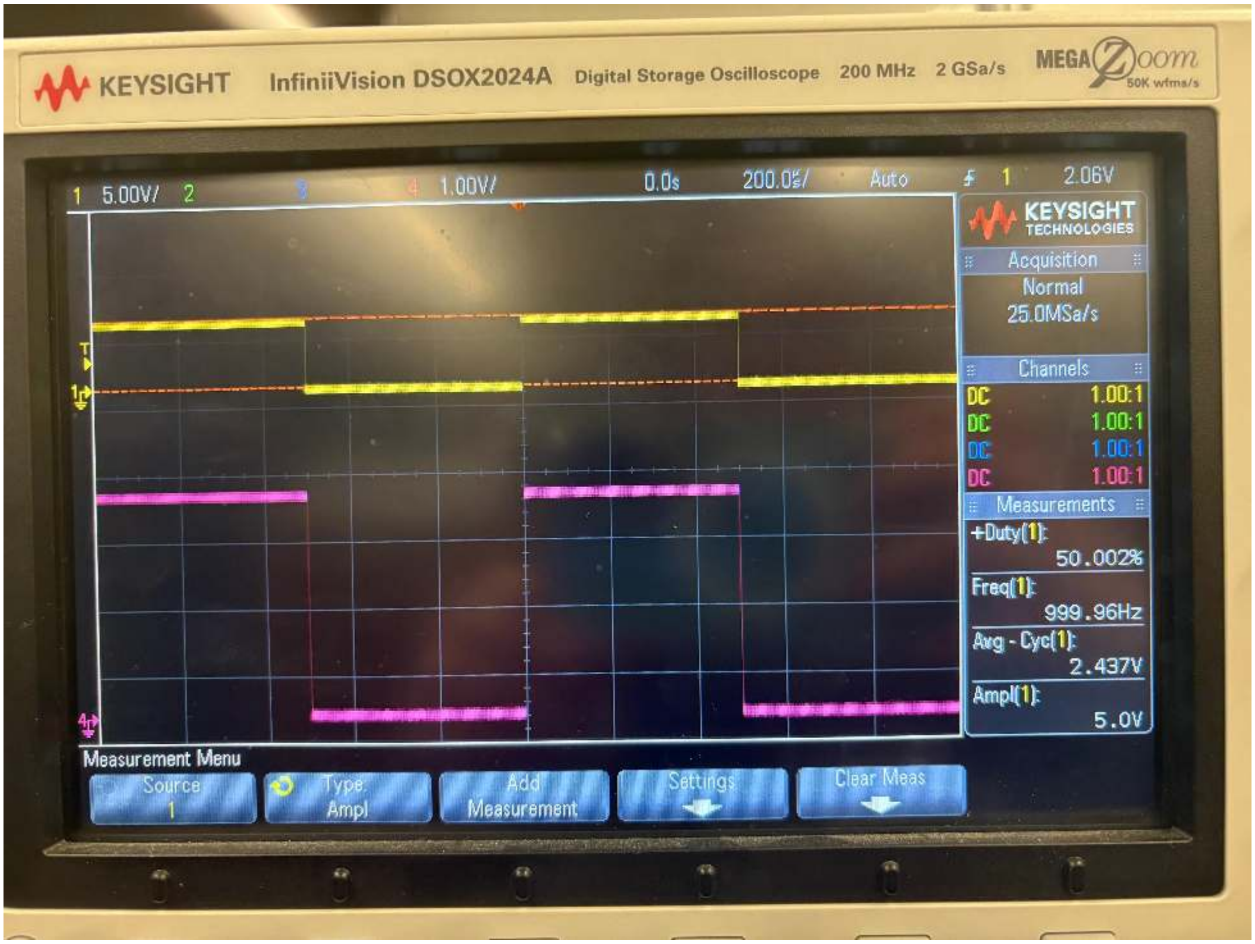
- View basic digital modulation signals on a scope
- Examine PWM signals generation and relate Duty Cycle to V average
- Examine a Sine PWM signal
- Use PWM to control the speed of a motor using an encoder

Part 1

IMG 2. *Predicted output of scope*



IMG 3. *Physical circuit*



IMG 4. *Scope output- yellow channel is output, pink channel is input.*

	Signal frequency	amplitude	Offset for signal frequency and amplitude	High level	Low level	Duty cycle
Pulse	1kHz	5Vpp	2.5 V	5V	0V	50%

Table 1. *Ideal settings, which we used.*

Frequency	Duty cycle	Average voltage
1.0 k Hz	50%	2.437

Table 2. *Scope measurements (also visible in image 4)*

We can calculate the average voltage as (Vmax- Vmin)* Duty cycle ratio. For the calculated average voltage in the table below, it is calculated as $(5v - 2.5v) \ * \ 0.5v \ = 2.5v$

Recorded average voltage	Vmax	Vmin	Duty cycle (as a ratio, in form duty%/ 100%)	Calculated average voltage
2.5V	5v	0v	50%	2.5V

Table 3.

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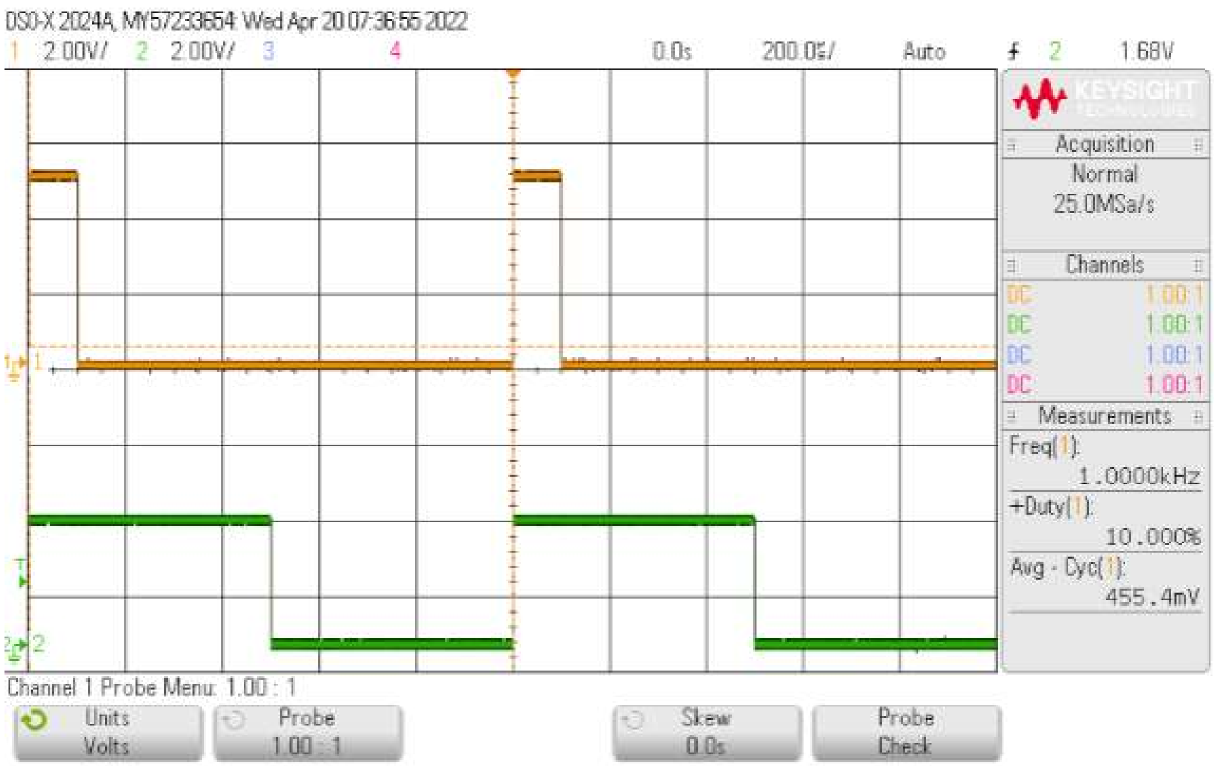
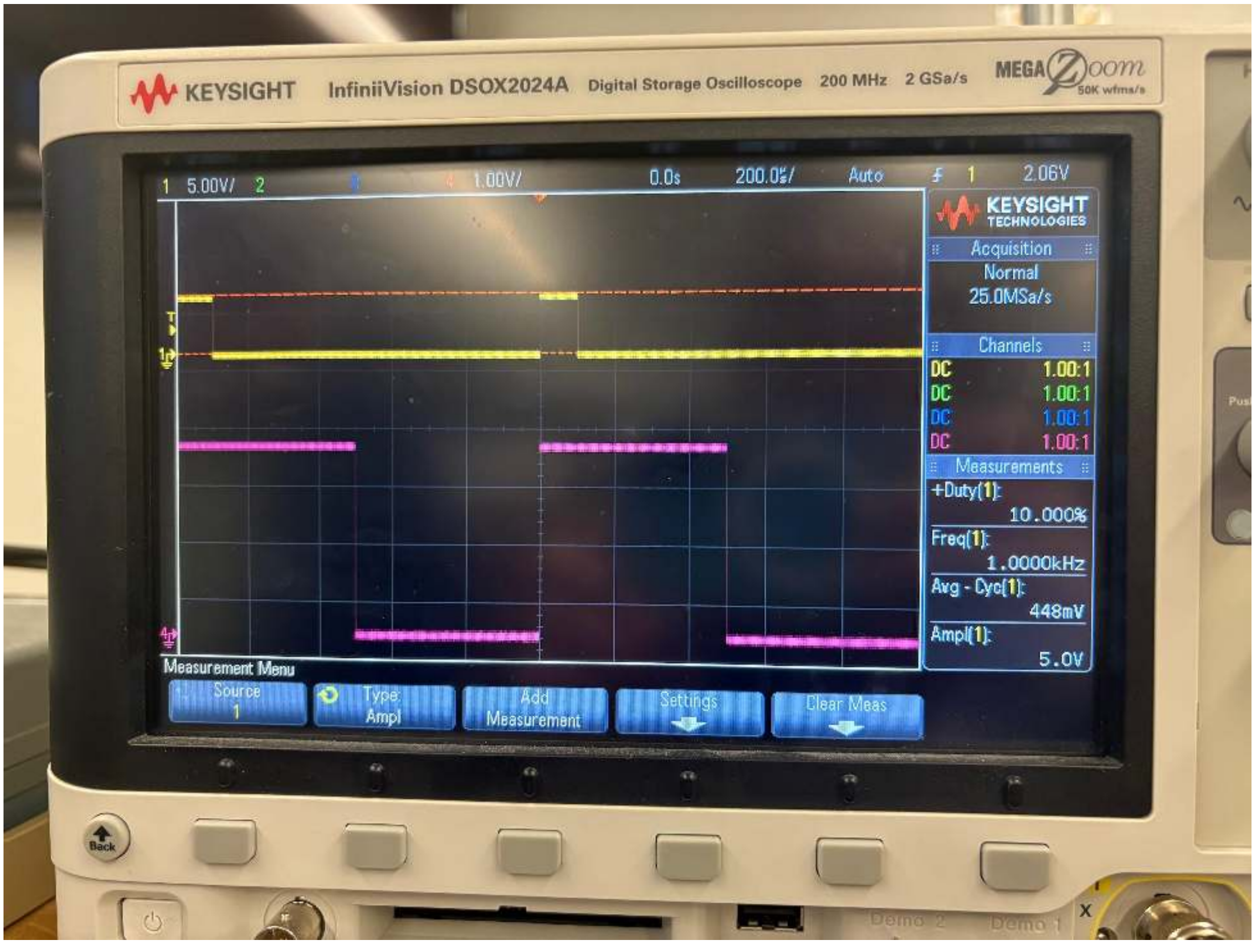
Then, the duty cycle was changed to 10% .



IMG 5. *Expected output on the scope*

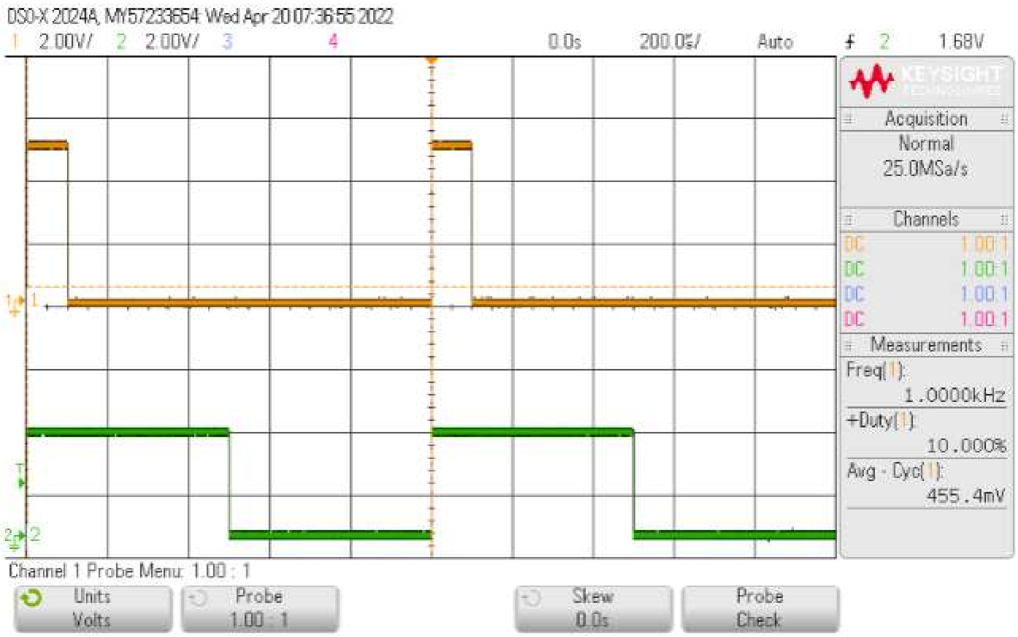


IMG 6. *Actual output on the scope.*

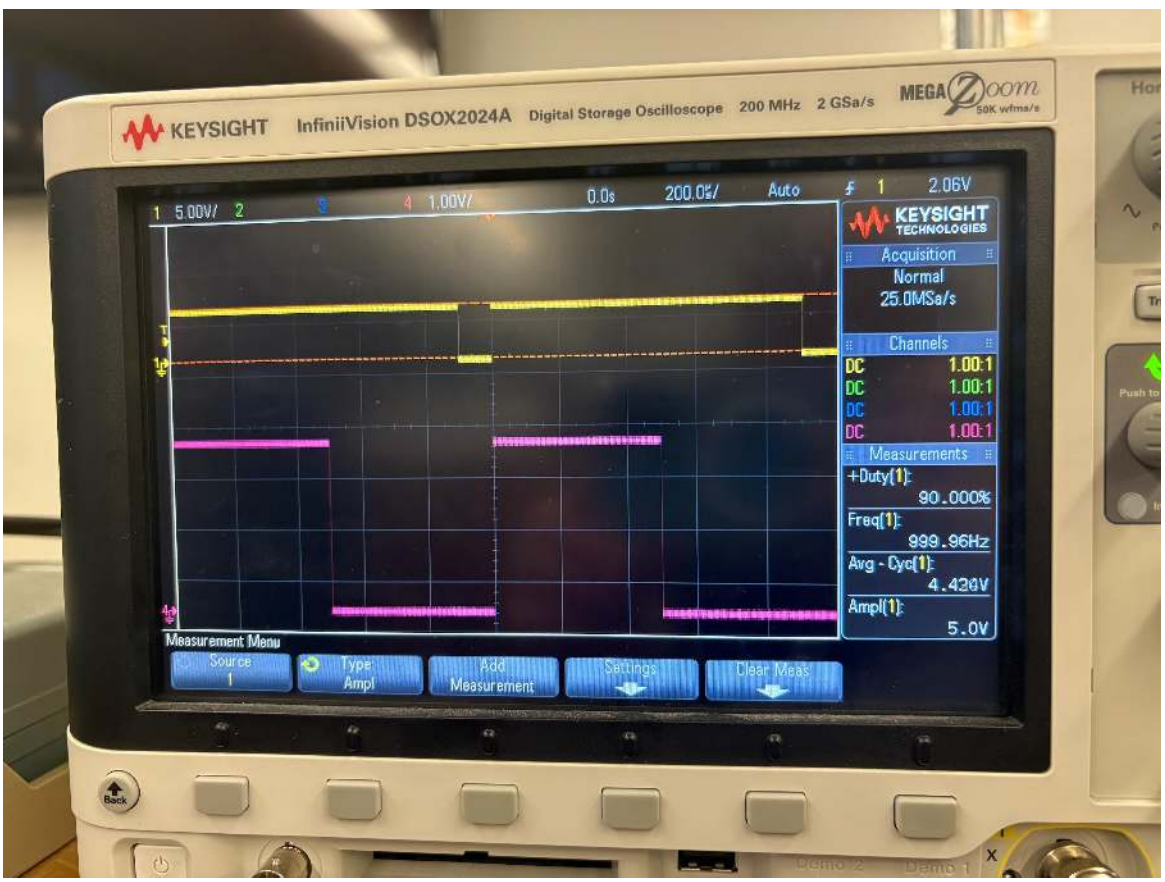
Duty cycle	Average voltage	Vmax	Vmin	Calculated results
10%	44.8mV	5v	0	50

Table 4.

Then, the duty cycle was changed to 90%



IMG 7. *Expected output*

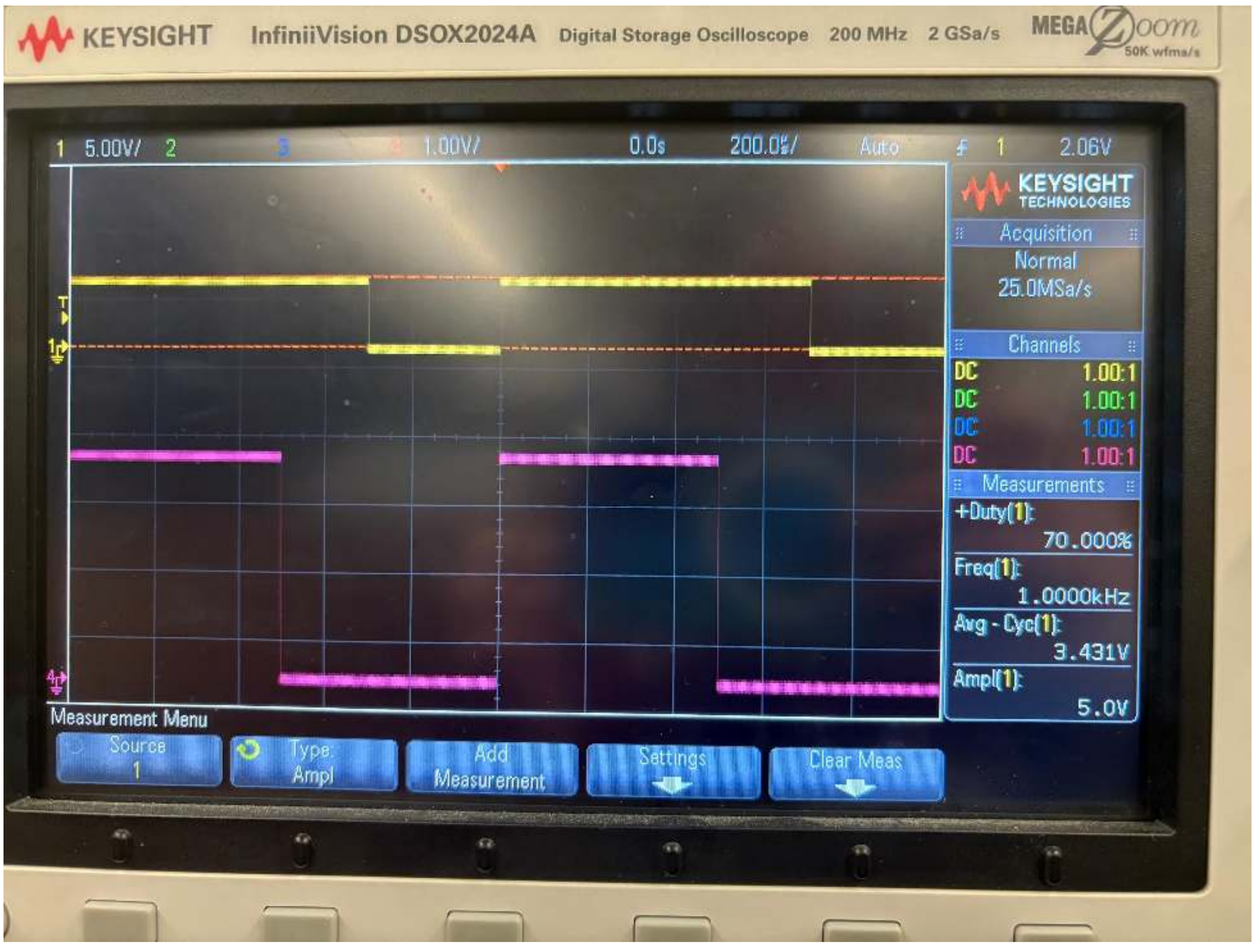


IMG 8. *Gathered output*

Duty cycle	Average voltage	Vmax	Vmin	Calculated results
90%	4.43	5v	0v	4.5v

Table 5.

Then, the duty cycle was changed to 70%

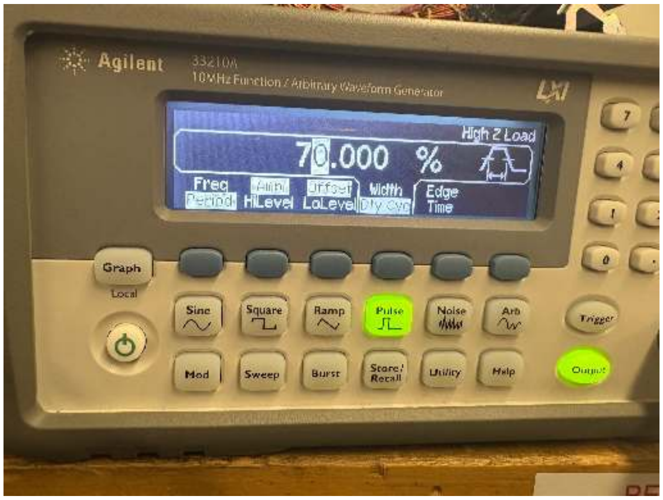


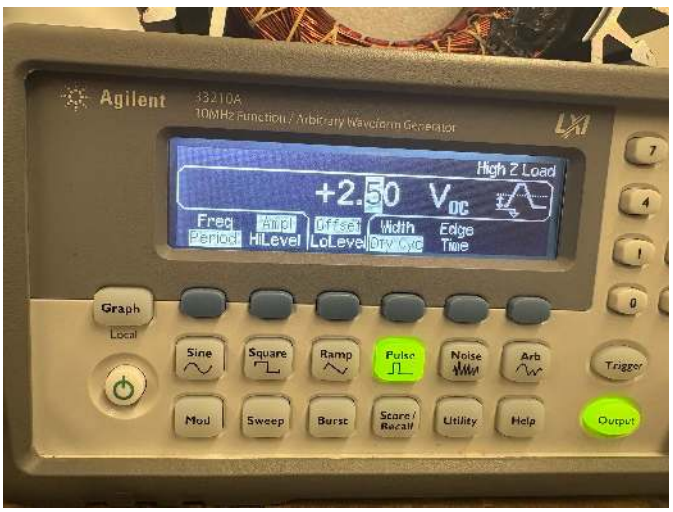
IMG 9. *Gathered output*

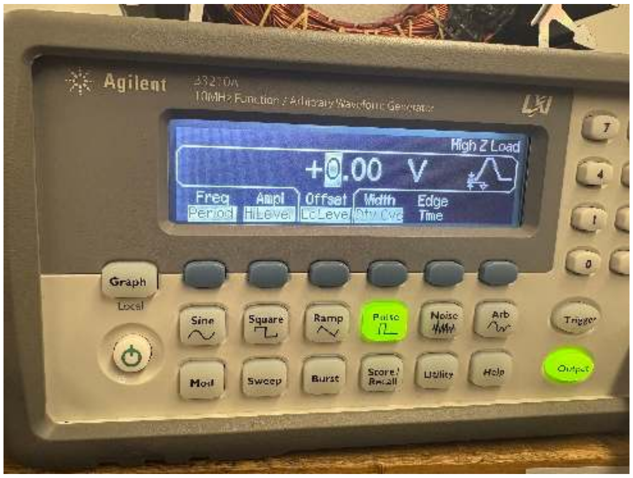
Duty cycle	Average voltage	Vmax	Vmin	Calculated results
70%	3.43	5v	0v	3.5

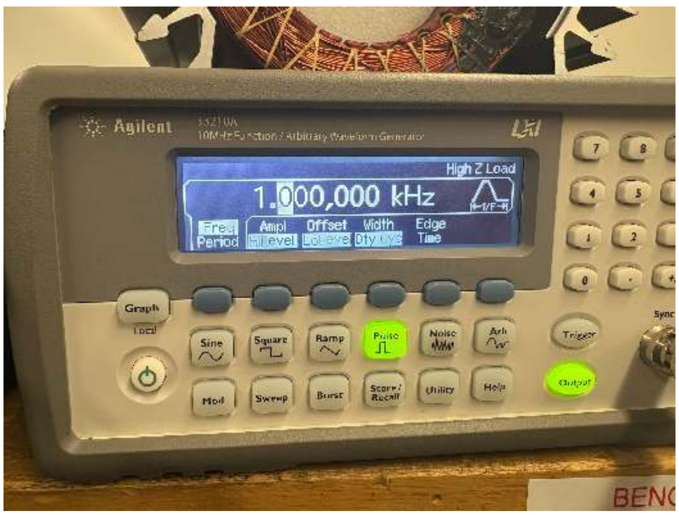
Table 6.

Here are the following settings needed for gathering the output in image 9 and in table 9:

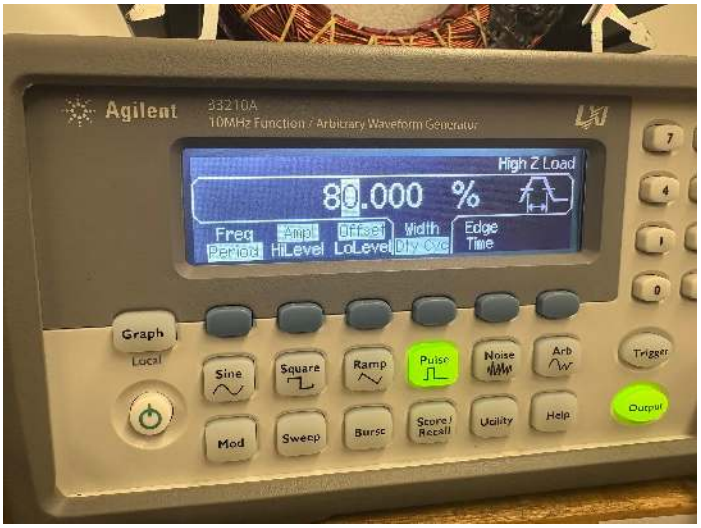






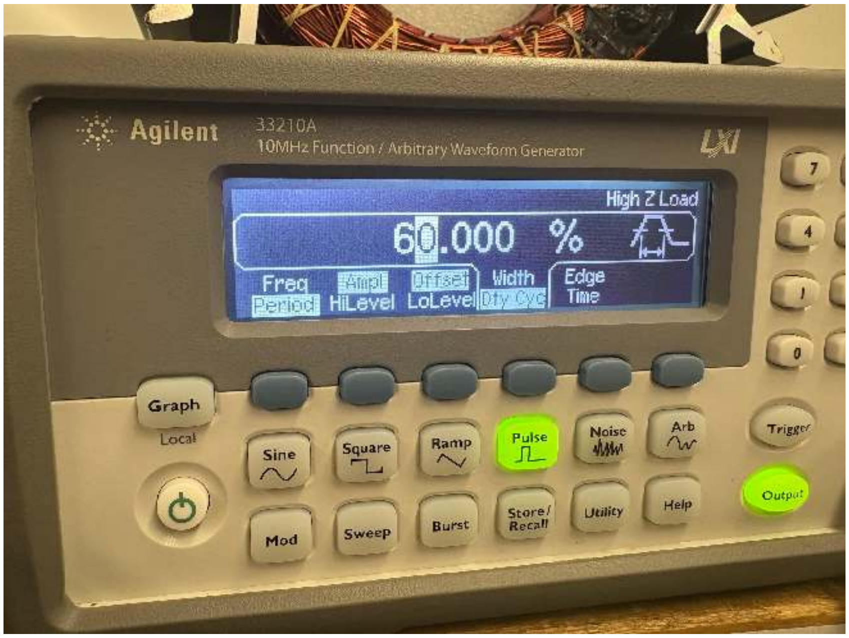


Here are some extra results- no calculations are needed, as the process is the same as previously explained.



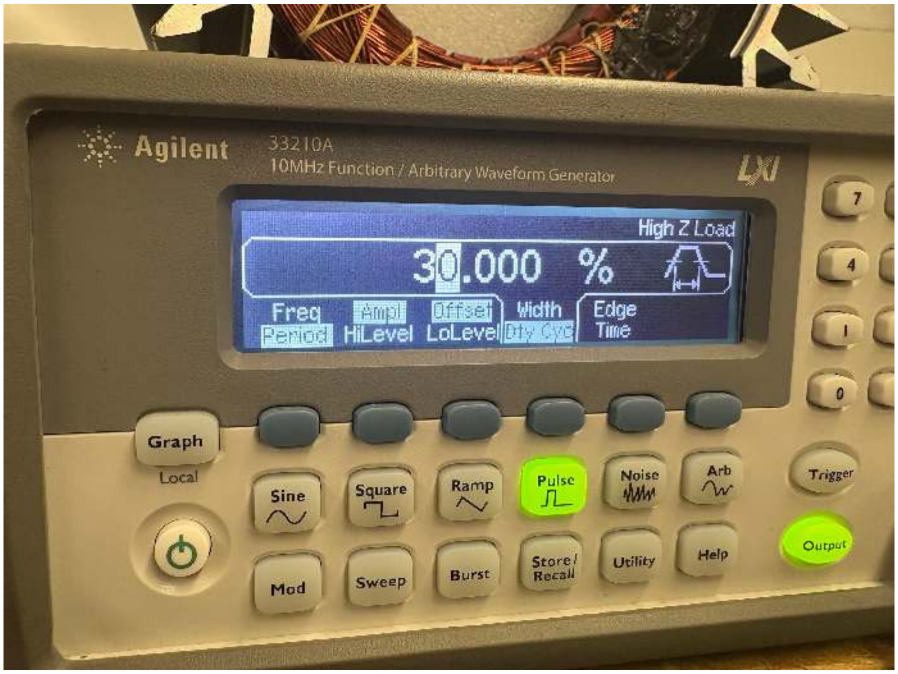
IMG 11. *80% duty cycle setting*

IMG 12. *80% duty cycle output*

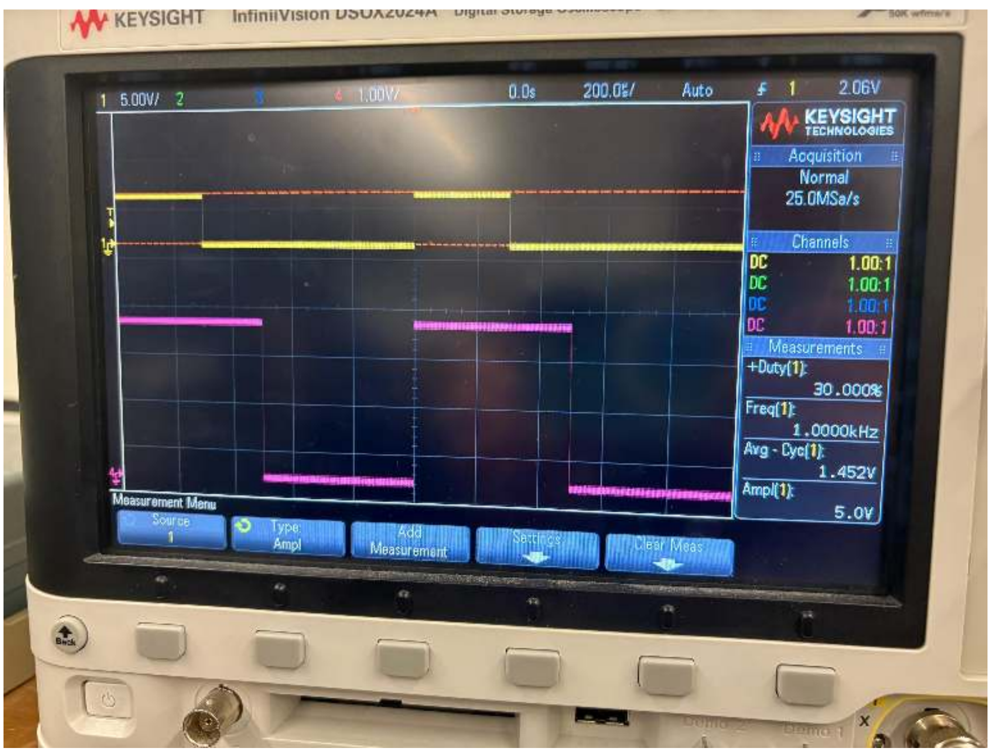


IMG 13. *60% duty cycle setting*

IMG 14. *60% duty cycle output*



IMG 15. *30% duty cycle setting*

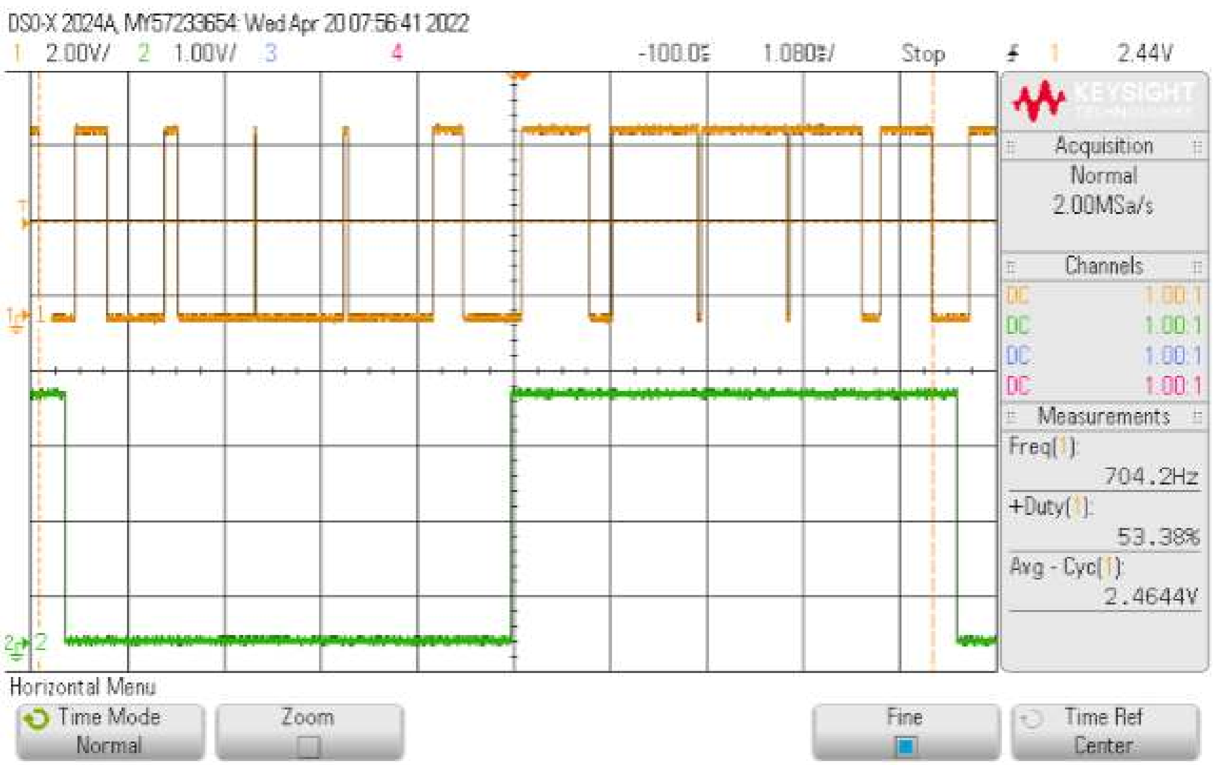


IMG 15. *30% duty cycle output*

Part 2

Part 2 is about sinusoidal modulated PWM. This is one of the most common PWM signals, where the pulse width or duty cycle is modulated (changed) by a sinusoidal signal as shown in the lab handout.

	Signal frequency	Low level	Duty cycle
Pulse	Approx 100Hz	Approx 0	50%



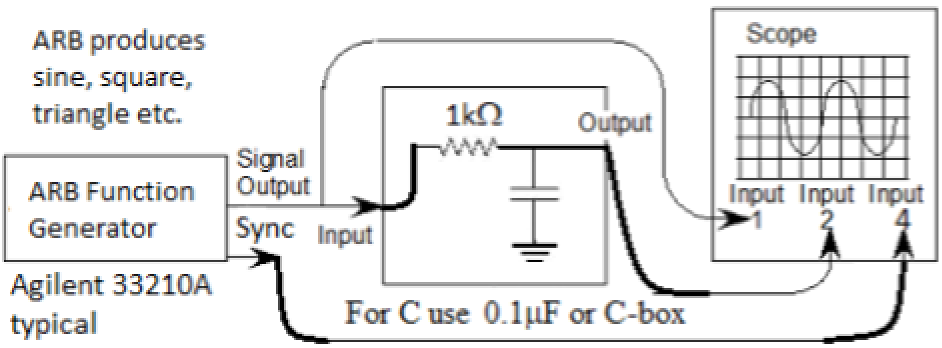
IMG 16. *What the scope should look like when we put the settings in and follow set up instructions from the lab.*

Part 2a- demodulating the sinusoidal modulated PWM

In order to view a modulation signal one must demodulate it by adding additional circuitry. In this case, an RC low pass filter.

Materials

- 1k ohm resistor
- 0.1 uF capacitor

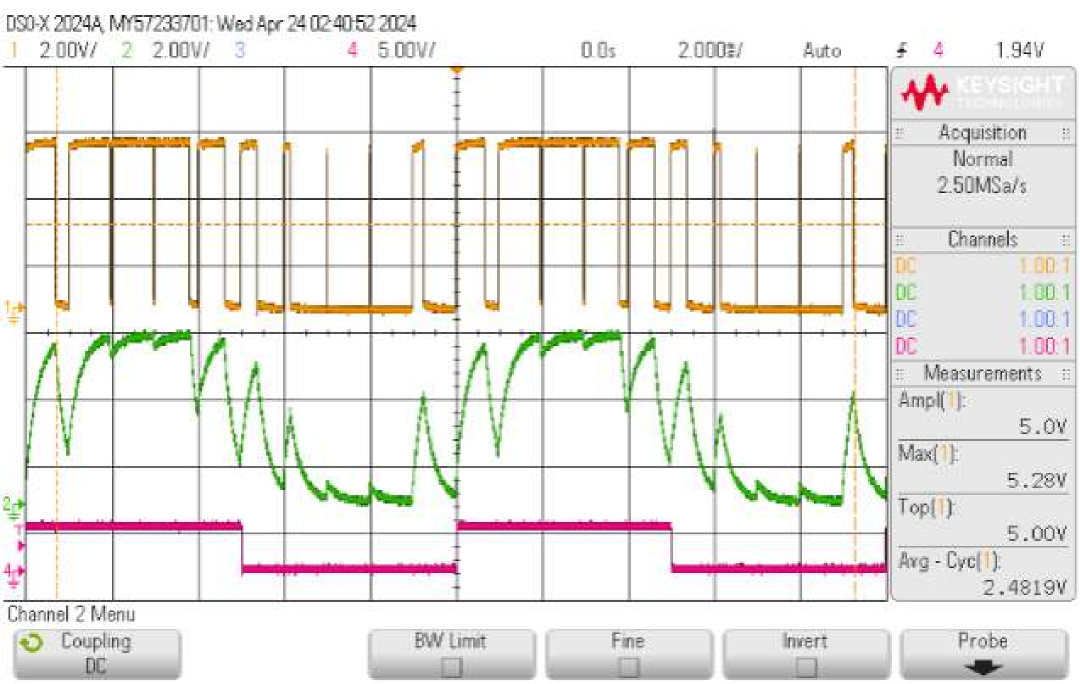


IMG 17. Schematic of the circuit

As one can see the demodulated signal (Ch2 green) is not very sinusoidal but if we increase the Capacitance to 0.22uFd (thereby lowering the filter frequency) to 723Hz it looks a little better but still not great.

IMG 18. *Expected results*

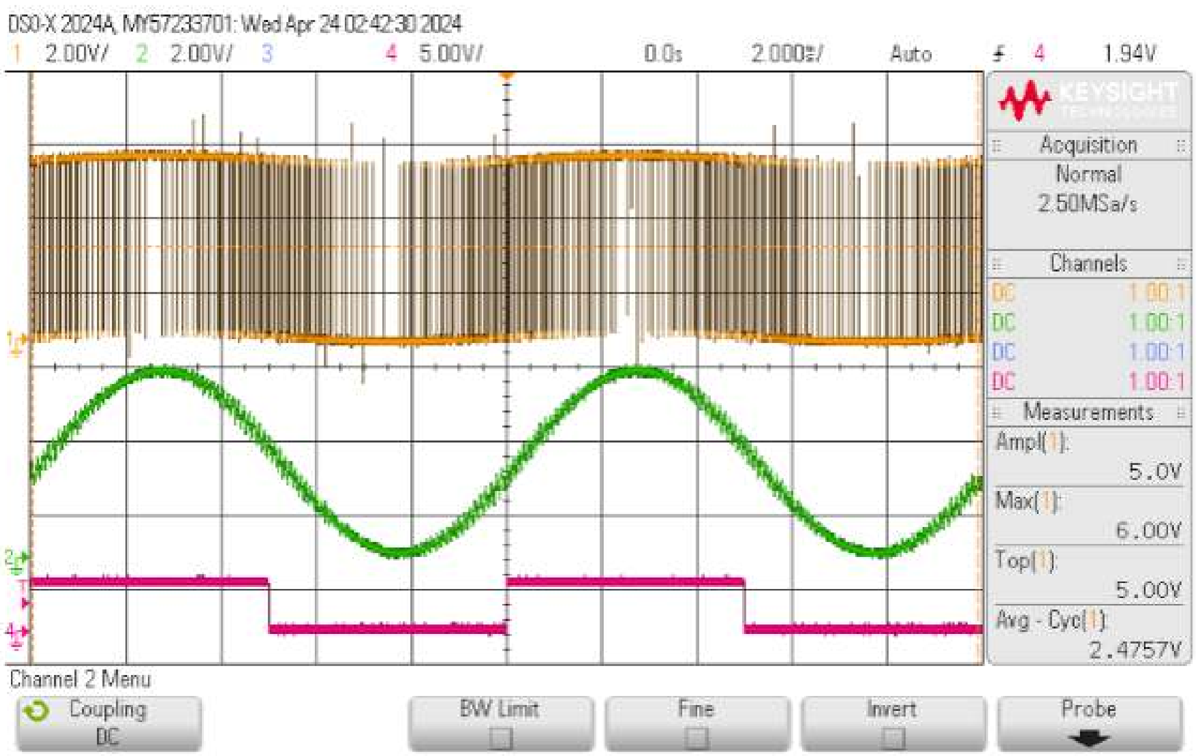
IMG 19 . *gathered results*



However, if we go back to the pulse frequency and change it to 10kHz we see a much better looking demodulated sine wave signal of 100Hz.

IMG 20. *Expected results of changing the pulse to 100hz*

IMG 21. *Gathered results*

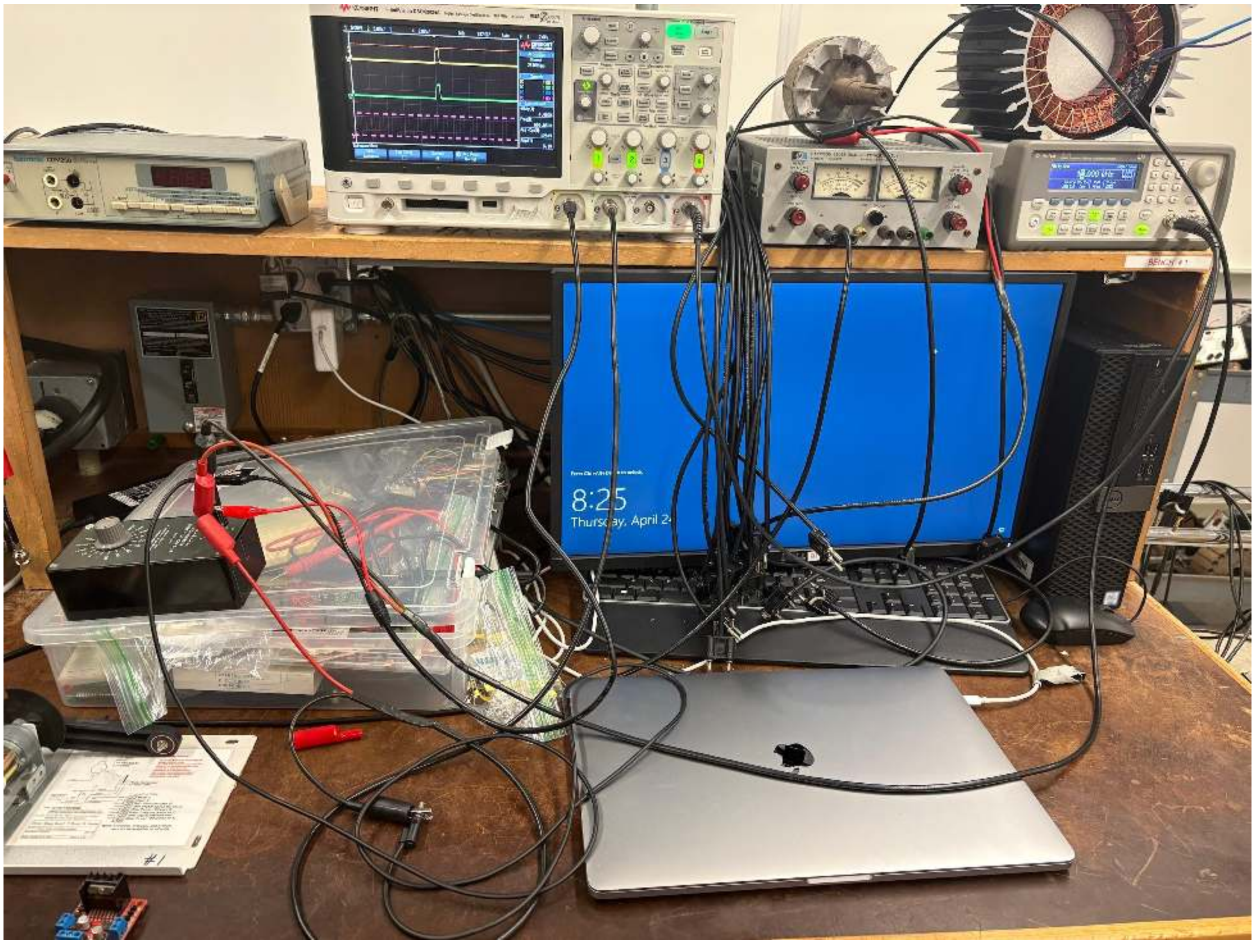


BTW an equivalent series LR low pass filter, L=R²C = 0.1H and measuring across R is has the same results.

IMG 22. *Expected results*

IMG 23. *Gathered results*

IMG 24. *Gathered results*

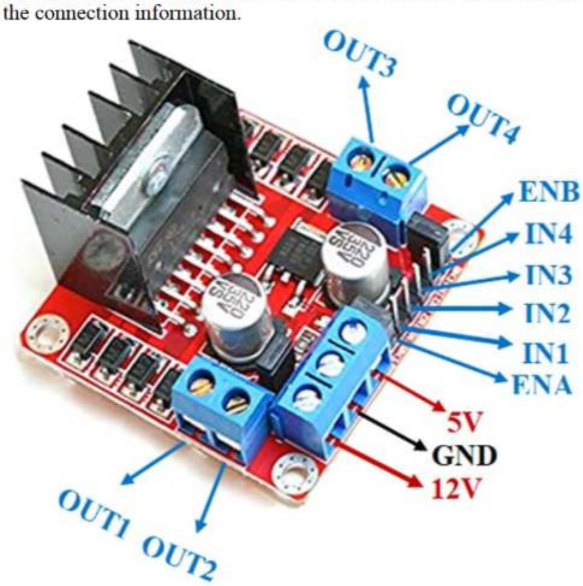


IMG 25.*Set up of scope*

Part 3

With DC motors, we know that adjusting the DC voltage controls the motor's speed; therefore, changing the PWM modulation percentage, which controls the average voltage (Vavg), will also control motor speed. The appropriate PWM frequency depends on the motor and other factors (see Electroboom video at https://youtu.be/yO9xIVv8ryc, time mark 12:30). In this case, the motor’s natural RL (resistor-inductor) low-pass filter behavior replaces the RC filter typically used to reduce high-frequency components. Since a sinusoidal signal can be embedded within a PWM signal, it is also possible to run AC-type motors using PWM frequency and an H-bridge, with voltage controlled by the modulation signal—something we will see later in Variable Frequency Drive (VFD) or Variable Speed Drive (VSD) systems. An alternative to building a circuit with discrete transistors is using an electronic H-bridge, such as the common L298D motor driver. The L298N module (see components101.com or https://lastminuteengineers.com/l298n-dc-stepper-driver-arduino-tutorial/) is typically set up for 12V but can support up to 35V; the 5V supply powers the logic and can come from either an external source or an onboard jumper-based regulator if the unit runs on 12V. This board is a dual motor driver, which is helpful for stepper motors, but in this lab, we will only use one output pair (OUT1/OUT2 or OUT3/OUT4). The IN1/IN2/IN3/IN4 pins control voltage polarity, and the ENA/ENB pins are the enable inputs to which a PWM signal can be connected.

to. See the connection information.



IMG 26. *Using outputs 1 and 2, 0 to 5 volts*

Video of it working: https://youtube.com/shorts/vTh002wul08?si=ZNyDJ4umqiO1bWfM

Conclusion

In this lab, we used Pulse Width Modulation (PWM) to control the voltage supplied to a DC motor, applying the formula to calculate the average voltage. This allowed us to control the motor's speed effectively. We also had to learn how to configure the function generator oscilloscope with the correct settings to properly observe and measure the PWM signal. In the final part of the lab, we used an H-bridge motor driver, specifically the L298N, to control both the speed and direction of the motor. This lab helped us understand the relationship between PWM, voltage control, and motor operation using real-world hardware.